DANIEL BROOKS

Senior Software Engineering | AI/ML | Data Engineering

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PROFESSIONAL SUMMARY

Freelance Software Engineer with 8+ years of experience in designing, developing, and delivering scalable, secure, and high-performance software solutions. Specializing in custom software development, ASP.NET and .NET applications, AI agents and automation, web and SaaS development, API and system integration, and cloud-based solutions. Experienced in building business applications, streamlining workflows, integrating intelligent technologies, and developing reliable digital products tailored to specific business needs. Passionate about solving complex technical challenges, improving operational efficiency, and transforming ideas into practical, innovative software solutions. Committed to delivering quality-driven engineering, maintainable applications, and technology solutions that help startups and established businesses scale, innovate, and achieve long-term growth.

TECHNICAL SKILLS

Programming
Python, C#, JavaScript, TypeScript, Java, Scala, Ruby, SQL/T-SQL

Backend & APIs
.NET / .NET Core, ASP.NET Core, Entity Framework Core, FastAPI, Flask, Django, Django REST Framework, Node.js, NestJS, Express, Ruby on Rails, REST APIs, GraphQL

Frontend & Mobile
HTML5, CSS3, JavaScript, TypeScript, React.js, Next.js, Angular, React Native, Flutter, Dart, Expo, Redux Toolkit, Tailwind CSS, Bootstrap, Vite, Responsive UI/UX, Android Development, iOS Development, Firebase, Push Notifications, REST API Integration, Mobile App Deployment, Google Play Store, Apple App Store

AI / ML / GenAI
LLM Integration, OpenAI API, Prompt Engineering, RAG, Embeddings, Vector Search, LangChain, LangGraph, LlamaIndex, AI Agents, MCP, Tool Calling, Hugging Face Transformers, PyTorch, TensorFlow, scikit-learn, Model Evaluation, AI Workflow Automation

Data Engineering & Analytics
ETL/ELT, batch and streaming pipelines, distributed ingestion, data modeling, validation, reconciliation, backfills, data quality, Apache Spark, Hadoop, pandas, NumPy, Kafka (familiar), Power BI

Databases & Storage
PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Firebase, Supabase, Cosmos DB, Pinecone, Qdrant, Weaviate, Chroma, pgvector, Prisma ORM, Drizzle ORM, AWS S3, Azure Blob Storage

Cloud & DevOps
Microsoft Azure, AWS, GCP, Azure Synapse, Azure AI Search, BigQuery, Dataflow, Dataproc (architecture familiarity), Docker, Kubernetes, AKS, Helm, Linux, Azure DevOps, GitHub Actions, CircleCI, YAML CI/CD

Testing & Operations
Unit testing, regression testing, API testing, monitoring, alerting, incident response, Sentry, Rollbar, AppSignal, logging, tracing, secure coding

Tools & Integrations
Git, GitHub, Jira, Slack, Gmail/Google APIs, Stripe, Twilio, OneSignal, CI/CD workflows

PROFESSIONAL EXPERIENCE

Senior Software Engineer & Team Lead 06/2025 – Present
AS-P Remote

- Developed scalable backend services, REST APIs, and custom software solutions using modern development frameworks and technologies.

- Integrated AI-powered APIs and Large Language Models (LLMs), implementing RAG workflows and evaluation processes to assess output quality, latency, and cost.
- Designed and optimized data pipelines, database operations, and system integrations to improve data quality, processing efficiency, and reliability.
- Implemented robust error handling, retries, circuit breakers, idempotency, logging, and monitoring to enhance application stability and system resilience.
- Supported automated testing, CI/CD pipelines, cloud deployments, and performance optimization to deliver secure, maintainable, and production-ready applications.

Senior Software Engineer 06/2022 – 05/2025
TheAMLab Remote

- Developed responsive web applications using React.js, Next.js, JavaScript, TypeScript, and modern frontend technologies.
- Built full-stack applications using the MERN stack, including MongoDB, Express.js, React.js, and Node.js.
- Developed cross-platform mobile applications using React Native and Expo, integrating REST APIs and Firebase services.
- Designed and integrated backend APIs, database solutions, and third-party services to deliver secure and scalable applications.
- Implemented testing, debugging, performance optimization, and CI/CD workflows to improve application reliability and deployment efficiency.

Senior Generative AI Engineer 02/2019 – 05/2022
Orthogonal Remote

- Developed AI-powered applications using Python, FastAPI, and modern backend technologies to deliver scalable and reliable solutions.
- Integrated Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), embeddings, and vector search for intelligent information retrieval.
- Built and optimized REST APIs, database integrations, and data processing workflows to improve application performance and reliability.
- Evaluated AI prompts and machine learning models using structured testing to assess output quality, accuracy, latency, and cost.
- Containerized applications using Docker and supported CI/CD workflows, cloud deployment, logging, and performance optimization.

Software Engineer, Machine Learning 05/2017 – 01/2019
Walmart Remote

- Developed predictive failure-detection models for retail POS devices using historical error logs, device performance data, and severity indicators.
- Processed and analyzed large datasets using DB2, Hadoop, Scala, and Python to identify failure patterns and improve predictive analytics.
- Developed and maintained C#/.NET services and SQL Server stored procedures to support data processing, system integration, and reliable application performance.
- Created Power BI reports and dashboards to visualize device failures, monitor error trends, and support data-driven maintenance decisions.

EDUCATION

Bachelor of Science in Computer Science (BSCS) 2018
University of the Punjab Lahore, Pakistan